Q1 JEE Main 2020 - 7 Jan (Morning)

Five numbers are in A.P., whose sum is 25 and product is 2520. If one of these five numbers is $-\frac{1}{2}$, then the greatest number amongst them is :

(A) $\frac{21}{2}$
(B) 16
(C) 5
(D) 7

Q2 JEE Main 2020 - 7 Jan (Morning)

The greatest positive integer k, for which 49^k+1 is a factor of the sum
$49^{125}+49^{124}+\ldots+49^2+49+1$, is

(A) 63
(B) 65
(C) 2
(D) 5

Q3 JEE Main 2020 - 7 Jan (Evening)

If the sum of the first 40 terms of the series, $3+4+8+9+13+14+18+19+\ldots..$ is $(102)m$, then m is equal to:

(A) 20
(B) 25
(C) 10
(D) 5

Q4 JEE Main 2020 - 7 Jan (Evening)

Let $a_1,\ a_2,\ a_3,\ldots$ be a G.P. such that $a_1<0, a_1+a_2=4$ and $a_3+a_4=16$. If $\sum_{i=1}^{9} a_i=4\lambda$, then λ is equal to :

(A) -513
(B) $-\frac{511}{3}$
(C) -171
(D) 171

Q5 JEE Main 2020 - 8 Jan (Morning)

Let $f : R \rightarrow R$ be such that for all $x \in R$ $\left(2^{1+x}+2^{1-x}\right)$, $f(x)$ and $\left(3^{x}+3^{-x}\right)$ are in A.P., then the minimum value of $f(x)$ is :

(A) 1
(B) 2
(C) 3
(D) 4

Q6 JEE Main 2020 - 8 Jan (Morning)

The sum $\sum\limits_{k=1}^{20}(1+2+3+\ldots+k)$ is

Q7 JEE Main 2020 - 8 Jan (Evening)

If the 10^{th} term of an A.P. is $\frac{1}{20}$ and its 20^{th} term is $\frac{1}{10}$, then the sum of its first 200 term is

(A) $201\frac{1}{2}$
(B) $101\frac{1}{2}$
(C) $301\frac{1}{2}$
(D) $100\frac{1}{2}$

Q8 JEE Main 2020 - 8 Jan (Evening)

The sum, $\sum\limits_{n=1}^{7}\frac{n(n+1)(2n+1)}{4}$ is equal to

Q9 JEE Main 2020 - 9 Jan (Morning)

Value of $\cos^{3}\frac{\pi}{8}\cos\frac{3\pi}{8}+\sin^{3}\frac{\pi}{8}\sin\frac{3\pi}{8}$ is

(A) $\frac{1}{2\sqrt{2}}$
(B) $\frac{1}{\sqrt{2}}$
(C) $\frac{1}{2}$
(D) $-\frac{1}{2}$

Q10 JEE Main 2020 - 9 Jan (Morning)

The product of
$2^{\frac{1}{4}}\cdot 4^{\frac{1}{16}}8^{\cdot\frac{1}{48}}\ldots\ldots\infty=$

(A) $\sqrt{2}$
(B) 2
(C) $2^{\frac{1}{4}}$
(D) 1

Q11 JEE Main 2020 - 9 Jan (Evening)

Let a_n be the n^{th} term of a G.P. of positive terms. If $\sum_{n=1}^{100} a_{2n+1} = 200$ and $\sum_{n=1}^{100} a_{2n} = 100$, then $\sum_{n=1}^{200} a_n$ is equal to:

(A) 175
(B) 150
(C) 300
(D) 225

Q12 JEE Main 2020 - 9 Jan (Evening)

If $x \sum_{n=0}^{\infty} (-1)^n \tan^{2n}\theta$ and $y = \sum_{n=0}^{\infty} \cos^{2n}\theta$, for $0 < \theta < \frac{\pi}{4}$, then:

(A) $y(1-x) = 1$
(B) $y(1+x) = 1$
(C) $x(1+y) = 1$
(D) $x(1-y) = 1$

Q13 JEE Main 2020 - 9 Jan (Evening)

The number of terms common the two A.P $3,\ 7,\ 11,\ \ldots.,\ 407$ and $2,\ 9,\ 16, \ldots.,\ 709$ is

Q14 JEE Main 2020 - 2 Sep (Morning)

If $|x| < 1, |y| < 1$ and $x \neq y$, then the sum to infinity of the following series $(x+y)+ \left(x^2+xy+y^2\right) + \left(x^3+x^2y+xy^2+y^3\right) + \ldots\ldots$ is :

(A) $\frac{x+y+xy}{(1+x)(1+y)}$
(B) $\frac{x+y-xy}{(1-x)(1-y)}$
(C) $\frac{x+y-xy}{(1+x)(1+y)}$

(D) $\frac{x+y+xy}{(1-x)(1-y)}$

Q15 JEE Main 2020 - 2 Sep (Evening)

If the sum of first 11 terms of an A.P., $a_1, a_2, a_3 \ldots$ is $0\left(a_1 \neq 0\right)$, then the sum of the A.P., a_1, a_3 $a_5, \ldots, a_{23}$ is $k a_1$, where k is equal to:

(A) $-\frac{121}{10}$
(B) $-\frac{72}{5}$
(C) $\frac{72}{5}$
(D) $\frac{121}{10}$

Q16 JEE Main 2020 - 2 Sep (Evening)

Let S be the sum of the first 9 terms of the series :
$\{x+k a\}+\left\{x^2+(k+2) a\right\}+\left\{x^3+(k+4) a\right\}+\left\{x^4+(k+6) a\right\}+\ldots$ where $a \neq 0$ and $x \neq 1$
If $S=\frac{x^{10}-x+45 a(x-1)}{x-1}$, then k is equal to :

(A) -3
(B) 1
(C) -5
(D) 3

Q17 JEE Main 2020 - 3 Sep (Morning)

If the first term of an A.P. is 3 and the sum of its first 25 terms is equal to the sum of its next 15 terms, then the common difference of this A.P. is :

(A) $\frac{1}{6}$
(B) $\frac{1}{4}$
(C) $\frac{1}{7}$
(D) $\frac{1}{5}$

Q18 JEE Main 2020 - 3 Sep (Morning)

The value of $(0.16)^{\log_{2.5}\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\ldots \text{ to } \infty\right)}$ is equal to

Q19 JEE Main 2020 - 3 Sep (Evening)

If the sum of the series $20+19\frac{3}{5}+19\frac{1}{5}+18\frac{4}{5}+\ldots$ upto n^{th} term is 488 and the n^{th} term is negative, then

(A) $n=41$
(B) n^{th} term is $-4\frac{2}{5}$
(C) $n=60$
(D) n^{th} term is -4

Q20 JEE Main 2020 - 3 Sep (Evening)

If m arithmetic means $(\mathrm{A.Ms})$ and three geometric means $(\mathrm{G.Ms})$ are inserted between 3 and 243 such that 4^{th} A.M. is equal to 2^{nd} G.M., then m is equal to

Q21 JEE Main 2020 - 4 Sep (Morning)

If $1+\left(1-2^2\cdot 1\right)+\left(1-4^2\cdot 3\right)+\left(1-6^2\cdot 5\right)+\ldots\ldots+\left(1-20^2\cdot 19\right)=\alpha-220\beta$, then an ordered pair (α,β) is equal to :

(A) $(10,103)$
(B) $(10,97)$
(C) $(11,97)$
(D) $(11,103)$

Q22 JEE Main 2020 - 4 Sep (Evening)

Let $a_1,a_2,\ldots.a_n$ be a given A.P. whose common difference is an integer and $\mathrm{S_n}=\mathrm{a_1}+\mathrm{a_2}+\ldots.+\mathrm{a_n}$ If $a_1=1,a_n=300$ and $15\leq n\leq 50$, then the ordered pair (S_{n-4},a_{n-4}) is equal to

(A) $(2490,249)$
(B) $(2480,249)$
(C) $(2490,248)$
(D) $(2480,248)$

Q23 JEE Main 2020 - 4 Sep (Evening)

The minimum value of $2^{\sin x}+2^{\cos x}$ is :

(A) $2^{-1+\sqrt{2}}$
(B) $2^{1-\sqrt{2}}$

(C) $2^{1-\frac{1}{\sqrt{2}}}$

(D) $2^{-1+\frac{1}{\sqrt{2}}}$

Q24 JEE Main 2020 - 5 Sep (Morning)

If $3^2 \sin 2\alpha - 1, 14$ and $3^{4-2} \sin 2\alpha$ are the first three terms of an A.P. for some α, then the sixth term of this A.P is :

(A) 65
(B) 78
(C) 81
(D) 66

Q25 JEE Main 2020 - 5 Sep (Morning)

If $2^{10} + 2^9 \cdot 3^1 + 2^8 \cdot 3^2 + \ldots . + 2 \cdot 3^9 + 3^{10} = S - 2^{11}$ then S is equal to :

(A) $2 \cdot 3^{11}$
(B) $3^{11} - 2^{12}$
(C) $\frac{3^{11}}{2} + 2^{10}$
(D) 3^{11}

Q26 JEE Main 2020 - 5 Sep (Evening)

If the sum of the second, third and fourth terms of a positive term G.P. is 3 and the sum of its sixth, seventh and.eighth terms is 243, then the sum of the first 50 terms of this G.P. is :

(A) $\frac{2}{13}\left(3^{50} - 1\right)$
(B) $\frac{1}{13}\left(3^{50} - 1\right)$
(C) $\frac{1}{26}\left(3^{49} - 1\right)$
(D) $\frac{1}{26}\left(3^{50} - 1\right)$

Q27 JEE Main 2020 - 5 Sep (Evening)

If the sum of the first 20 terms of the series $\log_{\left(7^{1/2}\right)} x + \log_{\left(7^{1/3}\right)} x + \log_{\left(7^{1/4}\right)} x + \ldots$ is 460 then x is equal to

(A) 7^2
(B) e^2
(C) $7^{1/2}$

(D) $7^{46/21}$

Q28 JEE Main 2020 - 6 Sep (Morning)

Let a, b, c, d and p be any non zero distinct real numbers such that $\left(a^2+b^2+c^2\right) p^2-2(ab+bc+cd)p+\left(b^2+c^2+d^2\right)=0$. Then :

(A) a, c, p are in G.P.
(B) a, b, c, d are in $A.P.$
(C) a, c, p are in A.P.
(D) a, b, c, d are in G.P.

Q29 JEE Main 2020 - 6 Sep (Evening)

The common difference of the A.P. $\mathbf{b}_1, \mathbf{b}_2, \ldots, b_m$ is 2 more than the common difference of A.P. $a_1, a_2, \ldots, a_n$, If $a_{40}=-159, a_{100}=-399$ and $\mathbf{b}_{100}=\mathbf{a}_{70}$, then $\mathbf{b}_1$ is equal to:

(A) -127
(B) -81
(C) 127
(D) 81

Answer Key

Q1 (B)	**Q2** (A)	**Q3** (A)	**Q4** (C)
Q5 (C)	**Q6** (1540)	**Q7** (D)	**Q8** (504)
Q9 (A)	**Q10** (A)	**Q11** (B)	**Q12** (A)
Q13 (14)	**Q14** (B)	**Q15** (B)	**Q16** (A)
Q17 (A)	**Q18** (4)	**Q19** (D)	**Q20** (39)
Q21 (D)	**Q22** (C)	**Q23** (C)	**Q24** (D)
Q25 (D)	**Q26** (D)	**Q27** (A)	**Q28** (D)
Q29 (B)			

Q1

Let terms be $a-2d,\ a-d,\ a,\ a+d,\ a+2d.$
sum $=25 \Rightarrow 5a=25 \Rightarrow a=5$
Product $=2520$
$(5-2d)(5-d)5(5+d)(5+2d)=2520$
$\Rightarrow \left(25-4d^2\right)\left(25-d^2\right)=504$
$\Rightarrow 625-100d^2-25d^2+4d^4=504$
$\Rightarrow 4d^4-125d^2+625-504=0$
$\Rightarrow 4d^4-125d^2+121=0$
$\Rightarrow 4d^4-121d^2-4d^2+121=0$
$\Rightarrow \left(d^2-1\right)\left(4d^2-121\right)=0$
$\Rightarrow d=\pm 1,\ d=\pm\frac{11}{2}$
$d=\pm 1,$ does not give $\frac{-1}{2}$ as a term
$d=\pm 1, \frac{-1}{2}$
$\therefore\ d=\frac{11}{2}$
$\therefore$ Largest term $=5+2d=5+11=16$

Q2

$\frac{(49)^{126}-1}{48}=\frac{\left((49)^{63}+1\right)\left(49^{63}-1\right)}{48}$

Q3

$S=\underbrace{3+4}+\underbrace{8+9}+\underbrace{13+14}+\underbrace{18+19}\ldots .40$ terms

$S=7+17+27+37+47+\ldots 20$ terms
$S_{40}=\frac{20}{2}[2\times 7+(19)10]$
$=10[14+190]=10[2040]=(102)(20)$
$\Rightarrow m=20$

Q4

$a_1 + a_2 = 4 \Rightarrow a_1 + a_1 r = 4 \quad \ldots \text{(i)}$

$a_3 + a_4 = 16 \Rightarrow a_1 r^2 + a_1 r^3 = 16 \quad \ldots \text{(ii)}$

$\frac{1}{r^2} + \frac{1}{4} \Rightarrow r^2 = 4$

$r = \pm 2$

$r = 2,\ a_1 (1+2) = 4 \Rightarrow a_1 = \frac{4}{3}$

$r = -2,\ a_1 (1-2) = 4 \Rightarrow a_1 = -4$

$$\sum_{i=1}^{a} a_i = \frac{a_1 (r^q - 1)}{r-1} = \frac{(-4)\left((-2)^9 - 1\right)}{-2-1} = \frac{4}{3}(-513) = 4\lambda$$

$\lambda = -171$

Q5

$$f(x) = \left(\frac{2^{1-x} + 2^{1+x} + 3^x + 3^{-x}}{2}\right)$$

Using $AM \geq GM$

$f(x) \geq 3$

Q6

$$= \sum_{k=1}^{20} \frac{k(k+1)}{2}$$

$$= \frac{1}{2} \sum_{k=1}^{20} \left(k^2 + k\right)$$

$$= \frac{1}{2}\left[\frac{20(21)(41)}{6} + \frac{20(21)}{2}\right]$$

$$= \frac{1}{2}\left[\frac{420 \times 41}{6} + \frac{20 \times 21}{2}\right]$$

$$= \frac{1}{2}[2870 + 210]$$

$= 1540$

Q7

$$T_{10} = \frac{1}{20} = a + 9d \qquad \ldots(\text{i})$$

$$T_{20} = \frac{1}{10} = a + 19d \qquad \ldots(\text{ii})$$

$$\Rightarrow \quad a = \frac{1}{200}, \; d = \frac{1}{200}$$

$$\Rightarrow \quad S_{200} = \frac{200}{2}\left[\frac{2}{200} + \frac{199}{200}\right] = \frac{201}{2} = 100\frac{1}{2}$$

Q8

$$\frac{1}{4}\left[\sum_{n=1}^{7}\left(2n^3 + 3n^2 + n\right)\right]$$

$$\frac{1}{4}\left[2\left(\frac{7.8}{2}\right)^2 + 3\left(\frac{7.8.15}{6}\right) + \frac{7.8}{2}\right]$$

$$\frac{1}{4}[2 \times 49 \times 16 + 28 \times 15 + 28]$$

$$\frac{1}{4}[1568 + 420 + 28] = 504$$

Q9

$$\cos^3\frac{\pi}{8}\left[4\cos^3\frac{\pi}{8} - 3\cos\frac{\pi}{8}\right] + \sin^3\left[3\sin\frac{\pi}{8} - 4\sin^3\frac{\pi}{8}\right]$$

$$= 4\cos^6\frac{\pi}{8} - 4\sin^6\frac{\pi}{8} - 3\cos^4\frac{\pi}{8} + 3\sin^4\frac{\pi}{8}$$

$$= 4\left[\left(\cos^2\frac{\pi}{8} - \sin^2\frac{\pi}{8}\right)\right]$$

$$\left[\left(\sin^4\frac{\pi}{8} + \cos^4\frac{\pi}{8} + \sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}\right)\right]$$

$$-3\left[\left(\cos^2\frac{\pi}{8} - \sin^2\frac{\pi}{8}\right)\right]$$

$$= \cos\frac{\pi}{4}\left[4\left(1 - \sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}\right) - 3\right] = \frac{1}{\sqrt{2}}\left[1 - \frac{1}{2}\right] = \frac{1}{2\sqrt{2}}$$

Q10

$2^{\frac{1}{4}+\frac{2}{16}+\frac{3}{48}+\ldots.\infty}$

$= 2^{\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots.\infty} = \sqrt{2}$

Q11

Let GP is $a,\ ar,\ ar^2\ldots.$

$$\sum_{n=1}^{100} a_{2n+1} = a_3 + a_5 + \ldots a_{201} = 200$$

$$\Rightarrow \frac{ar^2\left(r^{200}-1\right)}{r^2-1} = 200 \qquad \ldots(1)$$

$$\sum_{n=1}^{100} a_{2n} = a_2 + a_4 + \ldots a_{200} = 100 = \frac{ar\left(r^{200}-1\right)}{r^2-1} = 100\ \ldots(2)$$

Form (1) and (2) $r = 2$

add both

$$\Rightarrow a_2 + a_3 + \ldots.a_{200} + a_{201} = 300 \Rightarrow r\left(a_1 + \ldots a_{200}\right) = 300$$

$$\sum_{n=1}^{200} a_n = \frac{300}{r} = 150$$

Q12

$y = 1 + \cos^2\theta + \cos^4\theta + \ldots$

$$\Rightarrow y = \frac{1}{1-\cos^2\theta} \Rightarrow \frac{1}{y} = \sin^2\theta$$

$$x = \frac{1}{1-\left(-\tan^2\theta\right)} = \frac{1}{\sec^2\theta}$$

$$\Rightarrow \cos^2\theta = x$$

$$\Rightarrow \frac{1}{y} + x = 1$$

$$\Rightarrow y(1-x) = 1$$

Q13

First common terms $= 23$
common difference $= 7 \times 4 = 28$
Last term ≤ 407
$\Rightarrow \quad 23 + (n-1) \times 28 \leq 407$
$\Rightarrow \quad (n-1) \times 28 \leq 384$
$\Rightarrow \quad n \leq 13.71 + 1$
$n \leq 14.71$
So $n = 14$

Q14

$(x+y) + \left(x^2 + y^2 + xy\right) + \left(x^3 + x^2 y + x y^2 + y^3\right) + \cdots\cdots\cdots \infty$
$= \frac{1}{x-y}\left[\left(x^2 - y^2\right) + \left(x^3 - y^3\right) + \left(x^4 - y^4\right) + \ldots \infty\right]$
$= \frac{1}{x-y}\left[\frac{x^2}{1-x} - \frac{y^2}{1-y}\right] = \frac{(x-y)(x+y-xy)}{(x-y)(1-x)(1-y)}$
$= \frac{x+y-xy}{(1-x)(1-y)}$

Q15

Let common difference be d.
$\because a_1 + a_2 + a_3 + \ldots + a_{11} = 0$
$\therefore \quad \frac{11}{2}\{2a_1 + 10. d\} = 0$
$\therefore a_1 + 5d = 0$
$d = -\frac{a_1}{5}$
Now $a_1 + a_3 + a_5 + \ldots + a_{23}$
$= a_1 + (a_1 + 2d) + (a_1 + 4d) + \ldots + (a_1 + 22d)$
$= 12a_1 + 2d\frac{11 \times 12}{2}$
$= 12\left(a_1 + 11 \cdot -\frac{a_1}{5}\right)$
$= 12 \times \left(-\frac{6}{5}\right) a_1$
$= -\frac{72}{5} a_1$

Q16

Seires $(x + ka) + \left(x^2 + (k+2)a\right) + \ldots\ldots 9$ terms
$\Rightarrow \quad S = \left(x + x^2 + x^3 + \ldots\ldots 9 \text{ terms }\right) + a[k + (k+2)$
$+(k+4) + \ldots .9 \text{ terms }]$

$\Rightarrow \quad S=\frac{x\left(x^{9}-1\right)}{x-1}+\frac{9}{2}[2ak+8\times(+2a)]$

$\Rightarrow S=\frac{x^{10}-x}{x-1}+\frac{9ka+72a}{1}=\frac{x^{10}+45a(x-1)}{x-1}$

(given)

$\Rightarrow \frac{x^{10}-x+9a(k+8)(x-1)}{x-1}=\frac{x^{10}-x+45a(x-1)}{x-1}$

$\Rightarrow 9a(k+8)=45a$

$\Rightarrow k+8=5$

$\Rightarrow \quad k=-3$

Q17

Here $a=3\&s_{25}=s_{15}^{1}$

$\Rightarrow 2s_{25}=s_{40}$

$\Rightarrow 2\times\frac{25}{2}[6+24d]=\frac{40}{2}[6+39d]$

$\Rightarrow 25[6+24d]=20[6+39d]$

$\Rightarrow 150+600d=120+780\mathrm{d}$

$\Rightarrow 180\mathrm{d}=30$

$\Rightarrow d=\frac{1}{6}$

Q18

$\log_{2.5}\left(\frac{1}{3}+\frac{1}{3^{2}}+\frac{1}{3^{3}}+\ldots\infty\right)$

(0.16)

As sum of GP upto infinity $=\frac{a}{1-r}$

$\therefore \quad \frac{1}{3}+\frac{1}{3^{2}}+\frac{1}{3^{3}}+\ldots\infty=\frac{\frac{1}{3}}{1-\frac{1}{3}}=\frac{1}{2}$

$\therefore \quad (0.16)^{\log_{2.5}\left(\frac{1}{2}\right)}$

As $0.16=\frac{16}{100}=\left(\frac{4}{10}\right)^{2}=\left(\frac{10}{4}\right)^{-2}=(2.5)^{-2}$

$\therefore \quad (2.5)^{-2\log_{2.5}\left(\frac{1}{2}\right)}=\left(\frac{1}{2}\right)^{-2}=4$

Q19

$\mathrm{S}_{\mathrm{n}}=20+19\frac{3}{5}+19\frac{1}{5}+18\frac{4}{5}+\ldots\ldots\ldots$

$\because \mathrm{s}_{\mathrm{n}}=488$

$\frac{\mathrm{n}}{2}\left\{40+(\mathrm{n}-1)\cdot\left(-\frac{2}{5}\right)\right\}=488$

$\mathrm{n}\left(20-\frac{\mathrm{n}-1}{5}\right)=488$
$\mathrm{n}^2-101\mathrm{n}+2440=0$
$\therefore \quad (\mathrm{n}-40)(\mathrm{n}-61)=0$
For negative term $n=61$

$$n^{th} \text{ term } = T_{61} = 20+60\cdot\left(-\frac{2}{5}\right)$$
$$= -4$$

Q20

$3, A_1, A_2 \ldots A_m, 243$
As $243=3+(m+1)d$
$\Rightarrow \quad d=\frac{240}{(m+1)}$
Also $3, G_1, G_2, G_3, 243$
$\therefore$ As $G_2^2=243\times 3$
$G_2=\sqrt{243\times 3}=27$
Now $A_4=3+4d=3+\frac{960}{(m+1)}$
As $A_4=G_2$
$27=3+\frac{960}{m+1}$
$\therefore \quad m=39$

Q21

$1+\left(1-2^2.1\right)+\left(1-4^2.3\right)+\left(1-6^2.5\right)+\ldots\left(1-20^2.19\right)$
$S=1+\sum_{r=1}^{10}\left[1-(2r)^2(2r-1)\right]$
$=1+\sum_{r=1}^{10}\left(1-8r^3+4r^2\right)=1+10-\sum_{r=1}^{10}\left(8r^3-4r^2\right)$
$=11-8\left(\frac{10\times 11}{2}\right)^2+4\times\left(\frac{10\times 11\times 21}{6}\right)$
$=11-2\times(110)^2+4\times 55\times 7$
$=11-220(110-7)$
$=11-220\times 103=\alpha-220\beta \Rightarrow \alpha=11$
$\beta=103$
$\Rightarrow \quad (11,103)$

Q22

$a_1=1$ and $a_n=300$ and $d\in Z$
$300=1+(n-1)d$
$\Rightarrow d=\frac{299}{(n-1)}=\frac{23\times13}{(n-1)}$
$\therefore \mathrm{n}-1=13$ or 23 (as d is integer)
$\Rightarrow \mathrm{n}=14$ or 24
$\Rightarrow \mathrm{n}=24$ and $\mathrm{d}=13$
$a_{20}=1+19\cdot13=248$
$s_{20}=20\frac{(248+1)}{2}$
$=2490$

Q23

$\frac{2^{\sin x}+2^{\cos x}}{2}\geq\left(2^{\sin x+\cos x}\right)^{\frac{1}{2}}$
$\Rightarrow\quad 2^{\sin x}+2^{\cos x}\geq 2.2^{\frac{\sin x+\cos x}{2}}$
$\Rightarrow\quad 2^{\sin x}+2^{\cos x}\geq 2^{1-\frac{1}{\sqrt{2}}}$ (as $\sin x+\cos x\geq=\sqrt{2}\big)$

Q24

Given $3^{2\sin2\alpha-1}, 14, 3^{4-2\sin2\alpha}$ are in A.P.
So $3^{2\sin2\alpha-1}+3^{4-2\sin2\alpha}=28$
$\Rightarrow\frac{3^{2\sin2\alpha}}{3}+\frac{81}{3^{2\sin2\alpha}}=28$
$\Rightarrow\frac{t}{3}+\frac{81}{t}=28\quad\{$ Put $3^{2\sin2\alpha}=t\}$
$\Rightarrow t^2-84t+243=0\Rightarrow t=81, t=3$
$\Rightarrow$ When $t=81$ when $t=3$
$\Rightarrow\sin2\alpha=2(\text{ Not possible })2\sin2\alpha=1$
$\sin2\alpha=\frac{1}{2}$
$2\alpha=\frac{\pi}{6}$
$\alpha=\frac{\pi}{12}$
So, I term $a=3^{\circ}=1, d=14-1=13$
Now, $T_6=a+5d=1+65=66$

Q25

LHS is G.P of common ratio $\frac{3}{2}$

$\therefore \frac{2^{10}\left(1-\left(\frac{3}{2}\right)^{11}\right)}{\left(1-\frac{3}{2}\right)}=S-2^{11}$

$\Rightarrow 2^{10} \frac{\left(\frac{3^{11}-2^{11}}{2^{11}}\right)}{\frac{1}{2}}=S-2^{11}$

$\Rightarrow \mathrm{s}=3^{11}$

Q26

Let the first term be "a' and common ratio be 'r'.

$\therefore \quad \operatorname{ar}\left(1+r+r^{2}\right)=3$

and $\operatorname{ar}^{5}\left(1+r+r^{2}\right)=243$

From (i) and (ii),

$r^{4}=81 \Rightarrow r=3$ and $a=\frac{1}{13}$

$s_{50}=\frac{a\left(r^{50}-1\right)}{r-1}=\frac{3^{50}-1}{26}$

Q27

$S=\log_{7} x^{2}+\log_{7} x^{3}+\log_{7} x^{4}+\ldots\ldots 20$ terms

$\Rightarrow \log_{7}\left(x^{2} \cdot x^{3} \cdot x^{4} \cdot \ldots\ldots x^{21}\right)=460$ given

$\Rightarrow \log_{7} x^{(2+3+4 \ldots 21)}=460$

$\Rightarrow \quad(2+3+4+\ldots .+21) \log_{7} x=460$

$\Rightarrow \frac{20}{2}(2+21) \log_{7} x=460$

$\log_{7} x=\frac{460}{230}=2$

$x=7^{2}=49$

Q28

$\left(a^{2} p^{2}+2 a b p+b^{2}\right)+\left(b^{2} p^{2}+2 b c p+c^{2}\right)+\left(c^{2} p^{2}\right.$
$\left.+2 c d p+d^{2}\right)=0$

$\Rightarrow \quad(a p+b)^{2}+(b p+c)^{2}+(c p+d)^{2}=0$

$\therefore \quad a p+b=b p+c=c p+d=0$

$\Rightarrow \frac{b}{a}=\frac{c}{b}=\frac{d}{c} \quad \therefore a, b, c, d$ are in G.P.

Q29

Let common difference of series
$a_1, a_2, a_3, \ldots, a_n$ be d.
$\because a_{40} = a_1 + 39d = -159$
and $a_{100} = a_1 + 99d = -399 \ldots$ (ii)
From eqn. (ii) and (i)
$d = -4$ and $a_1 = -3$
Common difference of $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \ldots$ be (-2)
$\because \quad b_{100} = a_{70}$
$\therefore \quad b_1 + 99(-2) = (-3) + 69(-4)$
$\therefore \quad b_1 = 198 - 279$
$\therefore \quad b_1 = -81$